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AI Applications using PEAS Analysis

Application 1: AI-Based Smart Petrol Pump Assistant ⛽

Working -

● Detects the arriving vehicle using camera, number-plate recognition, or vehicle sensors.
● Suggests the appropriate fuel type based on vehicle information.
● Predicts maintenance needs using vehicle history and sensor data.
● Enables automatic payment through a linked digital payment method.

General PEAS Analysis -

PEAS Component	Detailed Description
Performance Measure	Accurate vehicle detection, correct fuel recommendation, useful maintenance prediction, fast service, secure payment, and customer satisfaction.
Environment	Petrol pump or fuel station where vehicles arrive, refuel, receive recommendations, and complete payment.
Actuators	Fuel dispensing interface, display, speakers, alerts, payment system, and recommendation interface.
Sensors	Camera, number-plate recognition, vehicle information, fuel dispenser sensors, payment input, service history, and vehicle sensor data.

10 Agent–Environment Combinations with Individual PEAS

Combination 1: Simple Reflex Agent + Fully Observable Environment

Selecting a fuel-related action when the required vehicle information is completely available.

PEAS Component	Detailed Description
Performance Measure	Correct vehicle identification, accurate fuel selection, fast response, and safe operation.
Environment	Fuel station where the vehicle, fuel requirement, and relevant vehicle information are clearly available. Fully observable for the task.
Actuators	Fuel dispenser, display, speakers, and payment interface.
Sensors	Camera, number-plate sensor, vehicle information input, and fuel dispenser sensor.

I observed that a simple reflex agent can directly connect a detected vehicle condition with an action when the required information is available.

Example: Vehicle detected → vehicle information identified → suitable fuel type displayed → customer starts refuelling.

Combination 2: Simple Reflex Agent + Static Environment

Performing basic predefined actions when the station setup and required rules remain unchanged.

PEAS Component	Detailed Description
Performance Measure	Reliable fuel selection, correct display, quick response, and consistent operation.
Environment	A fixed fuel station layout where the basic task and rules do not significantly change while the action is performed. Static for this task.
Actuators	Display, speakers, fuel dispenser, and payment interface.
Sensors	Vehicle detector, fuel dispenser sensor, and user input.

I observed that simple predefined petrol pump functions can work with direct rules without needing complex reasoning or historical data.
Example: Vehicle enters a designated lane → system detects the vehicle → available fuel options are displayed.

Combination 3: Model-Based Agent + Partially Observable Environment
Using previous vehicle information when the current vehicle state or sensor information is incomplete.

PEAS Component	Detailed Description
Performance Measure	Accurate vehicle-state estimation, improved fuel recommendation, and reliable maintenance advice.
Environment	Fuel station where camera angle, vehicle visibility, or sensor readings may be incomplete. Partially observable.
Actuators	Display, speakers, alerts, and recommendation interface.
Sensors	Camera, number-plate recognition, vehicle sensors, service history, and previous observations.

I observed that one observation may not reveal the complete vehicle condition. A model-based agent can use previous information as part of its internal state.
Example: Incomplete vehicle image + previous vehicle record → state estimation → improved fuel or maintenance recommendation.

Combination 4: Model-Based Agent + Sequential Environment
Tracking vehicle usage and service history across multiple visits.

PEAS Component	Detailed Description
Performance Measure	Correct tracking of vehicle history, meaningful comparisons, accurate maintenance prediction, and useful recommendations.
Environment	Fuel station where vehicles return over multiple visits and previous refuelling or maintenance information affects later decisions. Sequential.

Actuators	Display, alerts, payment interface, and maintenance recommendation interface.
Sensors	Vehicle identification, fuel dispenser data, service history, mileage input, and previous records.

I observed that maintenance prediction is not a one-time task. Earlier vehicle observations can influence how later conditions are interpreted.
Example: Previous visit data → current mileage/refuelling data → comparison → updated maintenance prediction.

Combination 5: Goal-Based Agent + Dynamic Environment
Selecting actions to achieve a fuel-service or vehicle-maintenance goal while vehicle and station conditions change.

PEAS Component	Detailed Description
Performance Measure	Achievement of the service goal, correct recommendations, efficient refuelling, safety, and customer satisfaction.
Environment	Fuel station where vehicle condition, queue length, fuel availability, traffic, and customer requirements can change. Dynamic.
Actuators	Display, fuel dispenser, alerts, payment interface, and recommendations.
Sensors	Vehicle sensors, camera, queue information, fuel availability data, and user input.

I observed that a goal-based agent is useful when the system has an objective rather than simply reacting to one input.
Example: Goal = safe and efficient refuelling → vehicle analysed → fuel/service requirement identified → suitable action selected.

Combination 6: Goal-Based Agent + Continuous Environment
Continuously monitoring vehicle and refuelling information against a safety or service goal.

PEAS Component	Detailed Description
Performance Measure	Accurate continuous monitoring, safe fuel dispensing, timely alerts, and progress toward the service goal.
Environment	Vehicle and fuel-dispensing conditions change continuously during the refuelling process. Continuous.
Actuators	Fuel dispenser, display, speakers, and safety alerts.
Sensors	Fuel-flow sensor, vehicle presence sensor, camera, and dispenser status sensor.

I observed that fuel dispensing changes continuously, so continuous monitoring is useful for detecting abnormal conditions quickly.

Example: Fuel dispensing starts → sensors continuously monitor flow and vehicle presence → abnormal condition detected → system provides an alert.

Combination 7: Utility-Based Agent + Stochastic Environment

Choosing a useful fuel or maintenance recommendation when vehicle needs and outcomes are uncertain.

PEAS Component	Detailed Description
Performance Measure	Quality and suitability of recommendations, safety, customer satisfaction, and reduced unsuitable advice.
Environment	Vehicle condition, fuel consumption, and maintenance outcomes are not perfectly predictable. Stochastic.
Actuators	Display, speakers, alerts, and recommendation interface.
Sensors	Vehicle sensor data, mileage, service history, fuel usage, and user input.

I observed that the same maintenance interval or recommendation may not suit every vehicle. A utility-based agent can compare several factors before selecting an option.

Example: AI evaluates mileage + usage + service history → estimates usefulness → recommends the more suitable maintenance action.

Combination 8: Utility-Based Agent + Dynamic Environment
Balancing cost, fuel needs, maintenance requirements, time, and changing station conditions.

PEAS Component	Detailed Description
Performance Measure	Good balance between cost, service quality, safety, adaptability, and customer satisfaction.
Environment	Changing fuel prices, vehicle conditions, fuel availability, queue length, and customer requirements. Dynamic.
Actuators	Display, payment interface, alerts, and recommendation system.
Sensors	Fuel price information, vehicle data, queue information, fuel availability, and user preferences.

I observed that the cheapest or fastest option is not automatically the best choice. The system should balance several competing factors.
Example: Low fuel level + long queue + maintenance alert → AI compares service factors → selects the most useful action.

Combination 9: Learning Agent + Continuous Environment
Learning vehicle usage patterns and preferences from repeated visits.

PEAS Component	Detailed Description
Performance Measure	Improved personalization, better maintenance prediction, accurate usage-pattern learning, and increased customer satisfaction.
Environment	Continuous interaction between vehicles and the smart petrol pump, with vehicle patterns developing over repeated visits. Continuous.

Actuators	Display, personalized recommendations, alerts, and payment interface.
Sensors	Vehicle identification, fuel history, mileage, service history, sensor data, and user feedback.

I observed that a learning agent can become more personalized because it learns from repeated refuelling and maintenance patterns.
Example: Vehicle regularly refuels at similar intervals → AI learns the pattern → future maintenance or fuel reminders become more personalized.

Combination 10: Learning Agent + Partially Observable Environment
Improving vehicle recommendations using previous records when complete current vehicle information is unavailable.

PEAS Component	Detailed Description
Performance Measure	Improved recommendations despite incomplete information, better maintenance prediction, and fewer unsuitable suggestions.
Environment	Fuel station where complete vehicle condition or service history may not be directly observable. Partially observable.
Actuators	Display, speakers, personalized alerts, payment interface, and recommendation system.
Sensors	Camera, vehicle identification, mileage, historical service data, fuel history, and user feedback.

I observed that the petrol pump cannot directly observe every part of a vehicle's condition. Learning from repeated records can improve future recommendations despite incomplete information.
Example: Incomplete current vehicle data + previous service history → AI learns patterns → future maintenance advice is improved.

4. Social Impacts

- Reduces waiting and improves fuel station service efficiency.
- Provides convenient and personalized fuel and maintenance recommendations.
- Supports faster digital and automatic payment.
- Encourages timely vehicle maintenance and better vehicle awareness.

5. Ethical Issues

- Privacy of vehicle and number-plate data.
- Security of automatic payment information.
- Bias or errors in fuel and maintenance recommendations.
- User consent and transparency about data collection and AI decisions.

Application 2: AI-Based Smart Restaurant Table Management □

Working -

● Predicts customer waiting time using reservations, queue information, and historical data.

● Allocates suitable tables according to party size and table availability.

● Optimizes seating to improve table utilization and reduce waiting.

● Updates table assignments as customers arrive, leave, or change their requirements.

General PEAS Analysis -

PEAS Component	**Detailed Description**
Performance Measure	Accurate waiting-time prediction, suitable table allocation, efficient seating, reduced waiting, quick response, and customer satisfaction.
Environment	Restaurant dining area where customers arrive, wait, receive table assignments, dine, and leave.
Actuators	Table allocation display, staff notifications, customer display, queue management interface, and reservation system.
Sensors	Table occupancy sensors, camera, reservation data, customer input, queue data, POS data, and historical restaurant records.

10 Agent–Environment Combinations with Individual PEAS

Combination 1: Simple Reflex Agent + Fully Observable Environment

Assigning a table when all relevant table and party information is completely available.

PEAS Component	Detailed Description
Performance Measure	Correct table allocation, fast response, reduced waiting, and accurate seating.
Environment	Restaurant where table occupancy, party size, and available tables are clearly known. Fully observable for the task.
Actuators	Table assignment display, staff notification, and customer notification interface.
Sensors	Table occupancy sensor, reservation data, customer input, and queue information.

I observed that a simple reflex agent can directly connect a known restaurant state with a table assignment when the required information is available.
Example: Party of four arrives → suitable four-seat table is available → system assigns the table → staff is notified.

Combination 2: Simple Reflex Agent + Static Environment
Handling basic seating rules when the restaurant layout and table information remain unchanged.

PEAS Component	Detailed Description
Performance Measure	Reliable table assignment, correct seating rules, and fast response.
Environment	A fixed restaurant layout where table positions and basic seating rules do not significantly change during the task. Static.
Actuators	Table display, staff notification, and customer notification.
Sensors	Table status, customer input, and reservation information.

I observed that basic table allocation can follow direct rules when the restaurant layout and seating rules are stable.

Example: Table becomes free → system detects the free table → next suitable waiting party is assigned.

Combination 3: Model-Based Agent + Partially Observable Environment
Using previous occupancy and queue information when the current restaurant state is incomplete.

PEAS Component	Detailed Description
Performance Measure	Accurate estimation of table availability, improved waiting-time prediction, and reliable allocation.
Environment	Restaurant where cameras may not capture every table and customer status may be incomplete. Partially observable.
Actuators	Customer display, staff alerts, and table allocation interface.
Sensors	Camera, occupancy sensors, reservation data, queue data, and previous observations.

I observed that one sensor may not reveal the complete restaurant state. A model-based agent can use previous observations to estimate the current state.
Example: Incomplete table view + previous occupancy information → state estimation → improved table allocation.

Combination 4: Model-Based Agent + Sequential Environment
Tracking table usage and customer flow across multiple seating events.

PEAS Component	Detailed Description
Performance Measure	Correct tracking of table history, meaningful queue comparisons, accurate waiting prediction, and suitable future allocation.
Environment	Restaurant where customers arrive, are seated, dine, and leave in sequence, and previous events affect later allocation. Sequential.
Actuators	Table allocation interface, staff notifications, and customer updates.

Sensors	Table status, reservation data, queue data, seating time, and historical records.

I observed that restaurant management is sequential because earlier seating events affect later waiting time and table availability.

Example: Customer seated → table occupancy recorded → customer leaves → table becomes available → next allocation is updated.

Combination 5: Goal-Based Agent + Dynamic Environment

Selecting seating actions to achieve low waiting time and efficient table utilization while restaurant conditions change.

PEAS Component	Detailed Description
Performance Measure	Achievement of low waiting time, high table utilization, correct allocation, and customer satisfaction.
Environment	Restaurant where arrivals, table availability, party sizes, reservations, and queue length can change. Dynamic.
Actuators	Table allocation display, staff alerts, customer notifications, and queue management system.
Sensors	Camera, table sensors, reservation system, queue data, customer input, and POS data.

I observed that a goal-based agent is useful when the restaurant has an objective such as reducing waiting time rather than simply assigning the first available table.

Example: Goal = reduce waiting → current queue and tables analysed → suitable table plan generated → customers are seated.

Combination 6: Goal-Based Agent + Continuous Environment

Continuously monitoring table occupancy and queue changes against a waiting-time goal.

PEAS Component	Detailed Description
Performance Measure	Accurate continuous monitoring, timely table assignment, reduced waiting, and efficient seating.

Environment	Customer arrivals, departures, table occupancy, and queue length change continuously. Continuous.
Actuators	Table display, staff alerts, customer notifications, and queue updates.
Sensors	Table occupancy sensors, camera, reservation data, and queue information.

I observed that restaurant occupancy can change within seconds, so continuous monitoring is useful for updating table assignments.

Example: Table becomes free → system detects the change → queue is checked → next suitable customer is notified.

Combination 7: Utility-Based Agent + Stochastic Environment

Choosing a table allocation that balances expected waiting time, customer comfort, and restaurant efficiency under uncertainty.

PEAS Component	Detailed Description
Performance Measure	Good seating decisions, reduced waiting, efficient utilization, customer comfort, and fewer unsuitable assignments.
Environment	Customer arrival time, dining duration, and table availability are not perfectly predictable. Stochastic.
Actuators	Table allocation system, staff notifications, and customer display.
Sensors	Reservation data, table status, queue data, historical dining duration, and customer input.

I observed that dining duration and arrival patterns are uncertain. A utility-based agent can compare possible seating choices instead of assuming everything will happen exactly as planned.

Example: Two tables are possible → AI considers party size + expected waiting + future demand → selects the higher-utility table.

Combination 8: Utility-Based Agent + Dynamic Environment

Balancing waiting time, table utilization, customer preferences, party size, and changing restaurant demand.

PEAS Component	Detailed Description
Performance Measure	Good balance between low waiting, high table utilization, customer comfort, adaptability, and satisfaction.
Environment	Changing customer arrivals, reservations, table availability, party sizes, and dining durations. Dynamic.
Actuators	Table allocation interface, staff alerts, customer notifications, and queue management.
Sensors	Table occupancy, reservations, queue data, customer preferences, and historical records.

I observed that the best table is not always the closest or largest table. The system should balance several competing factors.

Example: Large group arrives + limited tables + reservations approaching → AI compares options → assigns the table that gives better overall utility.

Combination 9: Learning Agent + Continuous Environment

Learning customer arrival patterns, table usage, and dining-duration patterns from repeated restaurant operations.

PEAS Component	Detailed Description
Performance Measure	Better waiting-time prediction, improved table allocation, accurate demand prediction, and long-term seating efficiency.
Environment	Restaurant operations occur continuously and patterns develop from repeated customer arrivals and table usage. Continuous.
Actuators	Table allocation system, personalized notifications, staff alerts, and queue recommendations.
Sensors	Reservation history, table occupancy, arrival times, dining duration, customer feedback, and POS data.

I observed that a learning agent can identify patterns such as busy periods, common party sizes, and typical dining durations.
Example: AI observes repeated Friday evening demand → learns the pattern → predicts higher waiting time and prepares better table allocation.

Combination 10: Learning Agent + Partially Observable Environment
Improving waiting-time and seating recommendations from historical data when the complete current restaurant state is unavailable.

PEAS Component	Detailed Description
Performance Measure	Better predictions from available data, improved personalization, efficient seating, and reduced unsuitable assignments.
Environment	Restaurant where some tables, customer intentions, or future arrivals are not directly observable. Partially observable.
Actuators	Table allocation interface, staff notifications, customer updates, and queue recommendations.
Sensors	Camera, table sensors, reservation history, queue data, POS data, customer feedback, and historical records.

I observed that the restaurant cannot directly observe every future arrival or exact dining duration. Learning from historical information can improve decisions despite incomplete information.
Example: Incomplete current occupancy + historical demand pattern → AI learns patterns → future waiting-time and seating recommendations are improved.

4. Social Impacts
- Reduces customer waiting time and improves dining experience.
- Improves table utilization and restaurant operational efficiency.
- Helps staff manage queues and seating more effectively.
- Provides more predictable and convenient restaurant service.

5. Ethical Issues

- Privacy of customers if cameras or occupancy monitoring are used.
- Fairness in table allocation and avoidance of customer discrimination.
- Security of reservation, customer, and payment-related data.
- Transparency about automated waiting-time predictions and seating decisions.